Project Overview

A machine learning mini project to predict death due to heart failure using 299 patient records.

Our Team

Name	Role	Entry no.
Rishit Jha	Execution lead	2026EEB1400
Kardile Pranav Parmeshwar	Video lead	2026MCB1534
Tappeta Sai Varsith Reddy	Documentation lead	2026AIB1086
Ujjwal Kumar Mahur	Presentation lead	2025DAB1286

Data Preparation

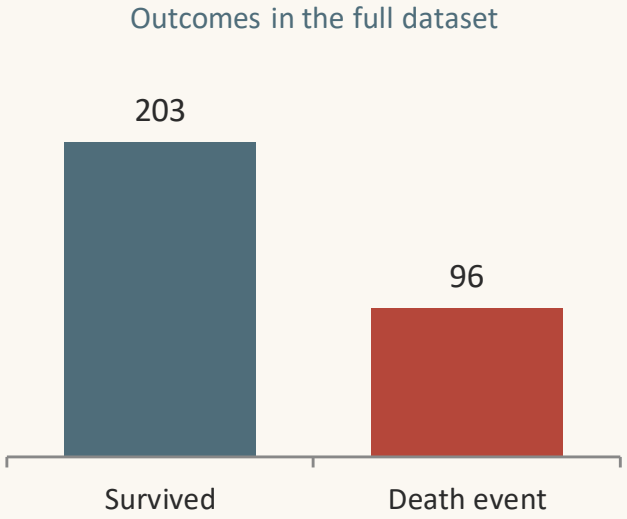
Load the CSV. 299 rows, 13 columns, no missing values and no duplicate rows.

Pick the target. DEATH_EVENT, where 0 means the patient survived and 1 means a death was recorded.

Pick the inputs. age, ejection_fraction, serum_creatinine, serum_sodium and time.

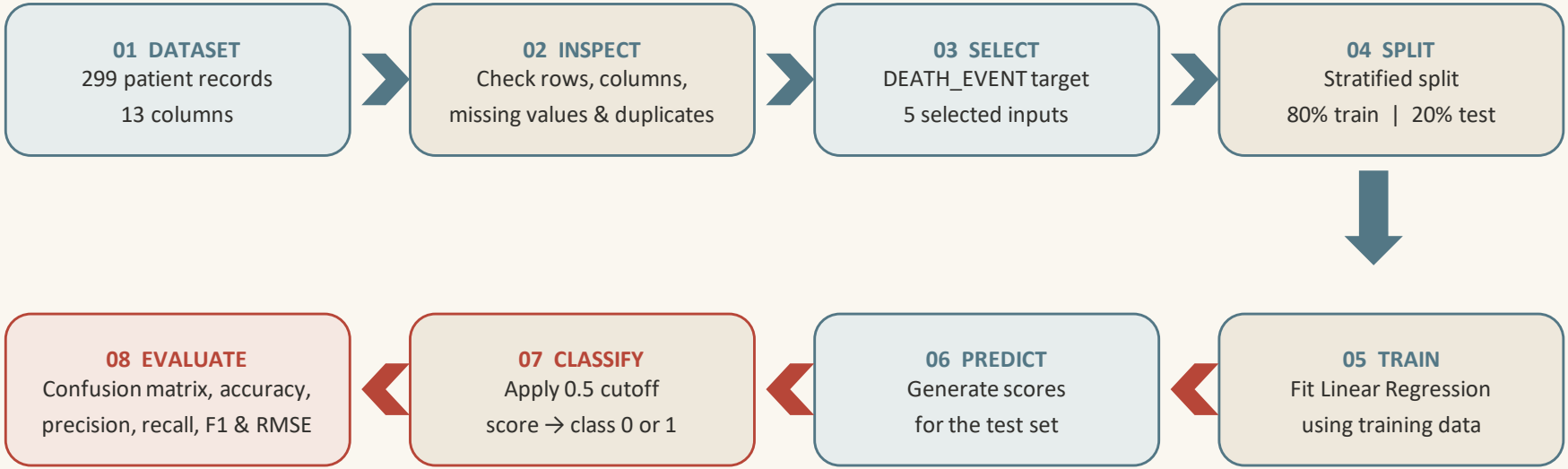
Split the data. 80% for training, 20% for testing, stratified so both parts keep the same survived/died ratio.

Keep the test set untouched. The model only ever sees the training rows while it is being fitted.



Project Pipeline

From clinical records to model evaluation



The test set is held out during training and used only for final evaluation.

Model Evaluation

How it works

We fit scikit-learn's LinearRegression on the training rows. It gives each test patient a score. If the score is 0.5 or more we call it a death (class 1), otherwise survived.

Inputs used

Age, ejection fraction, serum creatinine, serum sodium, follow-up time.

We also looked at MAE, RMSE and R^2, since the model is a regression underneath.

	Predicted survived	Predicted death
Actually survived	39	2
Actually died	8	11

83.3%	84.6%	57.9%	68.8%
accuracy	precision	recall	F1

Conclusion

We went through the whole workflow on one dataset: inspect the CSV, choose a target and five inputs, split 80:20, fit a linear regression, turn its scores into classes with a 0.5 cutoff, and evaluate on the held-out 60 patients.

It reached 83.3% accuracy but only 57.9% recall. A classification model would be the better next step.

Thank you.

299 patients

5 inputs used

83.3% test accuracy

57.9% test recall